# Week 2 — DDL: CREATE, ALTER, DROP, TRUNCATE, RENAME

**Lab:** Handout lab list **#2 — Data Definition Language** (CREATE, ALTER, DROP, TRUNCATE, RENAME) · lab manual **"Experiment 2: DDL COMMANDS"** (its index calls it Experiment 1) **Theory lectures this lines up with:** 12–17 (basic SQL: DDL)

Create all 13 tables with columns and data types only. Four things are created deliberately wrong so the ALTER / RENAME commands have real work to do. No keys, no defaults: that is next week.

## What was added

| File | What it is |
|---|---|
| `sql/01_create_tables.sql` | 13 `CREATE TABLE` statements: columns and types, nothing else. |
| `sql/02_alter_tables.sql` | `ADD COLUMN`, `RENAME COLUMN`, `MODIFY`, `DROP COLUMN`, `RENAME TABLE`, each fixing one deliberate flaw. |
| `demo/ddl_tour.sql` | Every DDL command from the manual on a throw-away copy (`CREATE TABLE … LIKE`), cleaned up afterwards. |
| `lab-practice/ddl_employee.sql` | **The lab manual's 9-question DDL assignment (EMPLOYEE table), solved in MySQL**, in its own `lab_practice` database. |

Files in `sql/` are **loaded by the build** (`scripts/build.sh`). Files in `queries/`, `demo/` and `lab-practice/` are run by hand and are checked by `scripts/check.sh`.

## What you should see

- 13 tables (`SHOW TABLES`). `DESCRIBE users` has `date_of_birth` right after `phone`, and the `Key` column is **empty** for every table.
- `devices.fingerprint_hash` is `varchar(64)`; `merchants.merchant_name` is `varchar(200)`; `fraud_alerts` is now called `alerts` and has no `notes` column.

## Concepts you must be able to explain (viva)

- **Data types.** `INT`, `BIGINT`, `DECIMAL(p,s)` (exact: use for money, never FLOAT), `CHAR` vs `VARCHAR`, `TEXT`, `ENUM`, `JSON`, `TIMESTAMP`, `BOOLEAN` (= `TINYINT(1)`).
- **ALTER TABLE.** `ADD`, `MODIFY` (change type/width), `RENAME COLUMN`, `DROP COLUMN`.
- **DELETE vs TRUNCATE vs DROP.** DELETE: chosen rows, can roll back, fires triggers. TRUNCATE: all rows, DDL, resets AUTO_INCREMENT, no rollback. DROP: rows *and* table.
- **DDL commits implicitly.** You cannot ROLLBACK a CREATE/ALTER/DROP/TRUNCATE.
- **Why bare tables first?.** A table without keys is legal but unsafe: it would accept duplicate users and orphan transactions. Week 3 shows the fix.

More questions with full answers: `week-11-final-polish/docs/viva_questions.md`. Syntax for the closed-book lab: `docs/LAB_EXAM_CHEATSHEET.md`.

## Handout step covered

Handout step 7 (SQL implementation: DDL).

# Results (captured from the running database)

## Evidence: Every DDL command on a throw-away table

Script: `week-02-ddl/demo/ddl_tour.sql`

```
+---------------+--------------+------+-----+-------------------+-------------------+
| Field         | Type         | Null | Key | Default           | Extra             |
+---------------+--------------+------+-----+-------------------+-------------------+
| user_id       | int          | NO   | PRI | NULL              | auto_increment    |
| full_name     | varchar(150) | NO   |     | NULL              |                   |
| email         | varchar(255) | NO   | UNI | NULL              |                   |
| phone         | varchar(20)  | YES  |     | NULL              |                   |
| date_of_birth | date         | YES  |     | NULL              |                   |
| created_at    | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+---------------+--------------+------+-----+-------------------+-------------------+
+---------------+--------------+------+-----+-------------------+-------------------+
| Field         | Type         | Null | Key | Default           | Extra             |
+---------------+--------------+------+-----+-------------------+-------------------+
| user_id       | int          | NO   | PRI | NULL              | auto_increment    |
| full_name     | varchar(150) | NO   |     | NULL              |                   |
| email         | varchar(255) | NO   | UNI | NULL              |                   |
| phone         | varchar(20)  | YES  |     | NULL              |                   |
| date_of_birth | date         | YES  |     | NULL              |                   |
| created_at    | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+---------------+--------------+------+-----+-------------------+-------------------+
+------------+-------------------------------------------------------------------------------------------------------------------------------------------------------
| Table      | Create Table
+------------+-------------------------------------------------------------------------------------------------------------------------------------------------------
| users_copy | CREATE TABLE `users_copy` (
  `user_id` int NOT NULL AUTO_INCREMENT,
  `full_name` varchar(150) COLLATE utf8mb4_unicode_ci NOT NULL,
  `email` varchar(255) COLLATE utf8mb4_unicode_ci NOT NULL,
  `phone` varchar(20) COLLATE utf8mb4_unicode_ci DEFAULT NULL,
  `date_of_birth` date DEFAULT NULL,
  `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (`user_id`),
  UNIQUE KEY `email` (`email`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci |
+------------+-------------------------------------------------------------------------------------------------------------------------------------------------------
+----------------------------+
| Tables_in_sentinel (users%) |
+----------------------------+
| users                      |
| users_backup               |
+----------------------------+
+----------------------------+
| Tables_in_sentinel (users%) |
+----------------------------+
| users                      |
+----------------------------+
```

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